Xin Wang

Ph.D. Student in Transportation Engineering | Trustworthy AI, Machine Learning & Optimization

Seattle, WA
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Homepage | *Google Scholar*

Education

2022–Present **Ph.D., Transportation Engineering**, *University of Washington*, Seattle, WA
Advisor: Prof. Xuegang (Jeff) Ban

2020–2022 **M.S., Applied Statistics**, *Renmin University of China*, Beijing, China

2016–2020 **B.S., Applied Mathematics**, *Central South University*, Changsha, China

Selected Publications

Constraint-Aware Influence Estimation in Deep Constrained Learning
Xin Wang, R. Tyrrell Rockafellar, Xuegang (Jeff) Ban.
Advances in Neural Information Processing Systems (NeurIPS), 2026.

Constrained Machine Unlearning of Learning-Based Intelligent Transportation Systems Models
Xin Wang, R. Tyrrell Rockafellar, Xuegang (Jeff) Ban.
Transportation Research Part C: Emerging Technologies, 190 (2026): 105763.

Transferability in Data Poisoning Attacks on Spatiotemporal Traffic Forecasting Models
Xin Wang, Feilong Wang, Yuan Hong, Xuegang (Jeff) Ban.
Transportation Research Part C: Emerging Technologies, 183 (2026): 105501.

Set-Valued Sensitivity Analysis of Deep Neural Networks
Xin Wang, Feilong Wang, Xuegang (Jeff) Ban.
Proceedings of the AAAI Conference on Artificial Intelligence (AAAI), (2025)

Data Poisoning Attacks on Traffic State Estimation and Prediction
Feilong Wang, Xin Wang, Yuan Hong, R. Tyrrell Rockafellar, Xuegang (Jeff) Ban.
Transportation Research Part C: Emerging Technologies, 168 (2024): 104577.

Data Poisoning Attacks in Intelligent Transportation Systems: A Survey
Feilong Wang, Xin Wang, Xuegang (Jeff) Ban.
Transportation Research Part C: Emerging Technologies, 165 (2024): 104750.

Infrastructure-Enabled Defense Methods against Data Poisoning Attacks on Traffic State Estimation and Prediction
Feilong Wang, Xin Wang, Xuegang (Jeff) Ban.
Conference on Emerging Technologies in Transportation Systems (TRC-30), 2025.

Research & Teaching Experience

2022–Present **Research Assistant**, *University of Washington*, Seattle, WA
- Develop machine learning and optimization methods for trustworthy learning, including sensitivity analysis, machine unlearning, robustness, and cybersecurity.
- Study transportation systems as a major application domain, including traffic forecasting, state estimation, and data-driven intelligent transportation systems.

Autumn 2025 **Teaching Assistant**, *CET 513: Optimization in Transportation, UW CEE*
- Led lab sections, held office hours, and graded assignments.

Autumn 2024 **Teaching Assistant**, *CET 513: Optimization in Transportation, UW CEE*
- Led lab sections, held office hours, and graded assignments.

Industry Experience

Jan–May 2021 **Machine Learning Engineer Intern**, *Baidu Inc.*, Beijing, China
- Worked on multi-objective ranking optimization for online video search using Pareto-efficient learning-to-rank methods.
- Optimized ranking quality and click-through objectives and identified Pareto-efficient solutions using NSGA-II.

Invited Talks & Presentations

Jan 2025 **Data Poisoning Attacks on Traffic State Estimation and Prediction**
ISTTT25

June 2024 **A Review of Data Poisoning Attacks in Intelligent Transportation Systems**
TRB 2025

Academic Service

Reviewer Transportation Research Part C: Emerging Technologies; TRB Annual Meeting; ICML; AAAI; NeurIPS